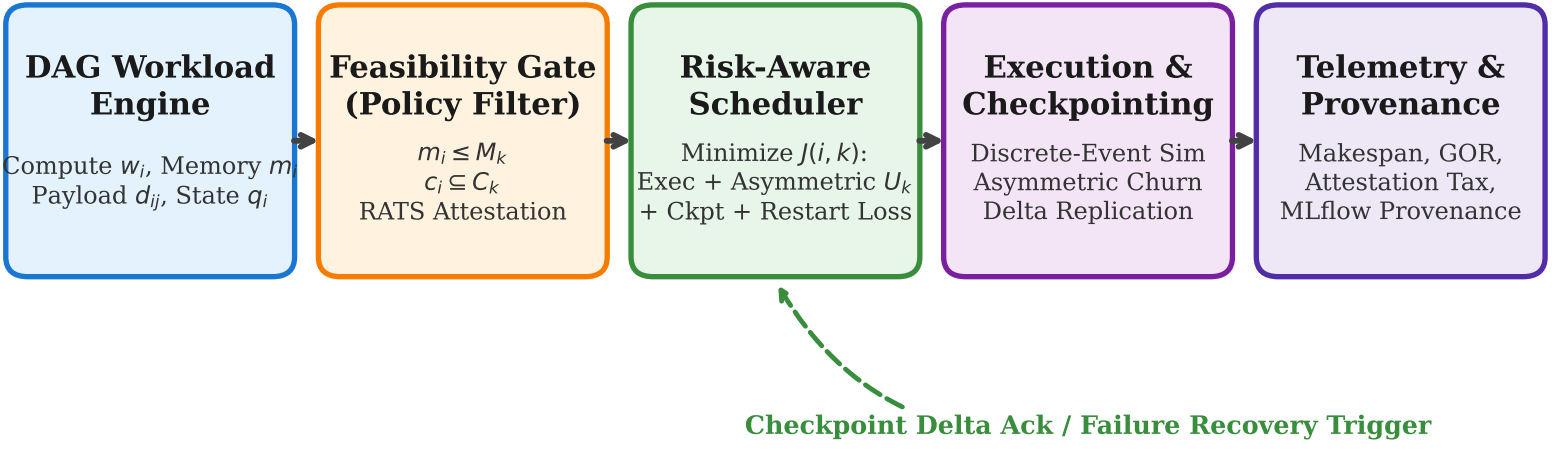GhostInTheMachine: End-to-End Execution Pipeline & Runtime Architecture